

Entertainment OS

# Universal Bootloader

Global generic solution

June 2024



# Contents

A decorative blue line graphic that starts as a horizontal line to the right of the title, then turns 90 degrees downward and continues as a vertical line.

- 4 Current build
- 6 Proposed Global Solution
- 8 The Goal
- 10 Competitors
- 16 Concept Design
- 20 Next steps

# Bootloader

The startup bootloader is a crucial component that bridges the gap between powering on the TV and loading the full operating system, ensuring that the device starts correctly and securely.

Bootloaders tend to initialise a minimal user interface or provide visual feedback (such as a logo or progress bar) to indicate that the TV is booting up.

# Current Build

What we have live today



# Current Build

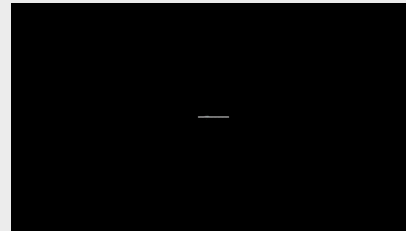
What we have live today

sky



- Results in dedicated Sky manufacturing line and Bootloader builds
- Not scalable to the range of partners we have

Hubbl



- Limited customer feedback in boot process
- Xumo requested a different approach

xumo



- Interim approach until new “universal” screen agreed
- Not scalable to the language options we have (Rogers example: English & French Canadian)

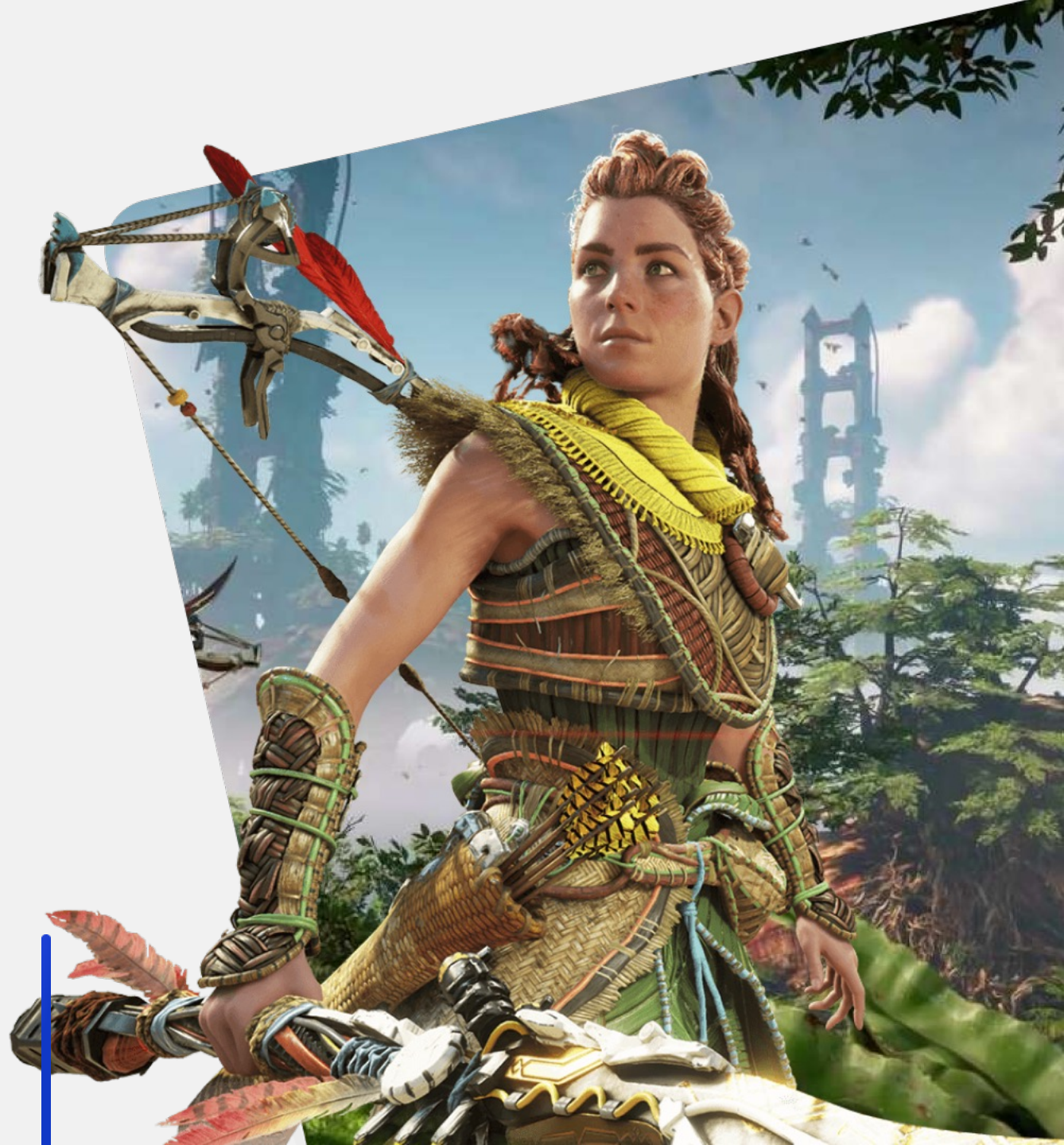
# Proposed Global solution





# The Goal

What, why and how



# The Goal

What are we trying to solve?

## Problem

The new proposed Universal EntOS brand poses a risk of brand encroachment with partners.

## Why is it important?

1. **Critical path for Rogers launch**, it would turn the programme amber.
2. **Reduce number of Software builds and variants** creating manufacturing, cost & testing efficiencies.
3. **Allows us to scale** out to territories and partners easily.

## How do we solve it?

A new generic, Global solution that either does not lean on the EntOS brand at all or does so in a covert way.

# Competitors

What do our competitors do?



# Apple TV

Bootloader



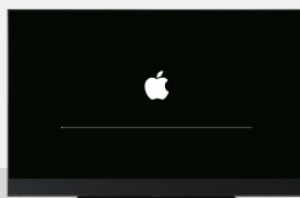
Initial Boot Screen  
16s



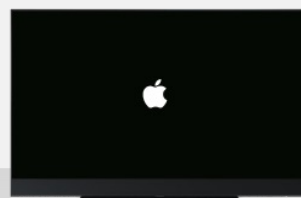
Static logo  
1s



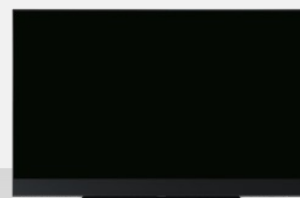
Black  
10s



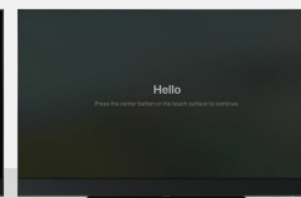
Logo & progress bar  
7s



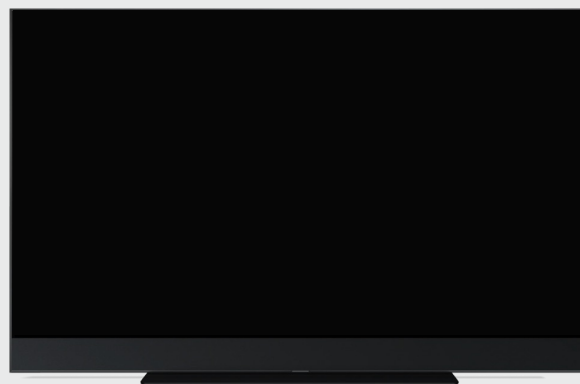
Static logo  
1s



Black  
4s



Setup & Activation



Video

# Apple iOS (Phone iPad)

Bootloader



Initial Boot Screen  
13s



Middleware Boot Screen  
5s



Setup & activation

# Roku

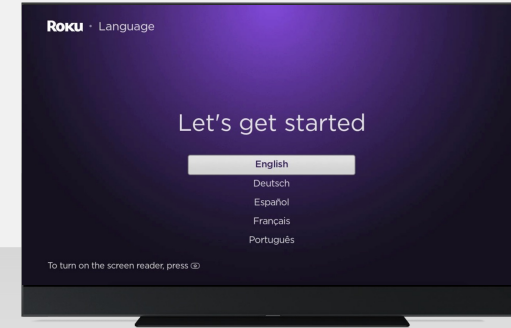
Bootloader



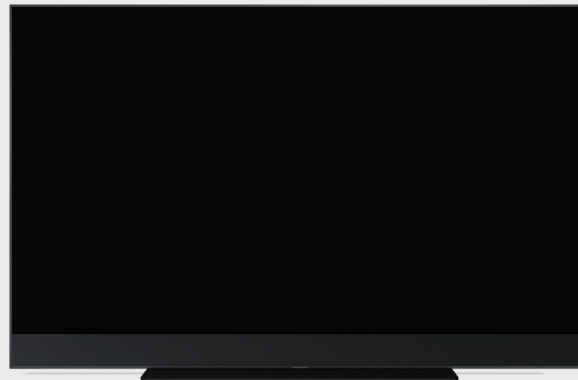
Initial Boot Screen  
5s



Logo animation (loop)  
20s



Setup & Activation



Video

# Google TV

Bootloader



Black Screen  
10s



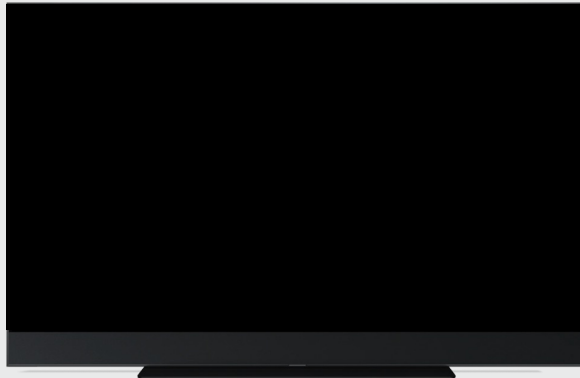
Static logo  
20s



Logo animation  
10s



Black Screen  
3s



Video

# Fire TV

## Bootloader



Black Screen  
5s



Static logo  
2s



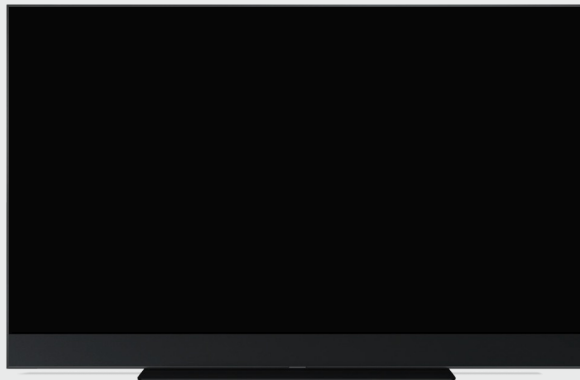
Static status message  
6s



Static logo  
7s



Logo animation  
3 minutes



Video

# Concept design



# Concept Design

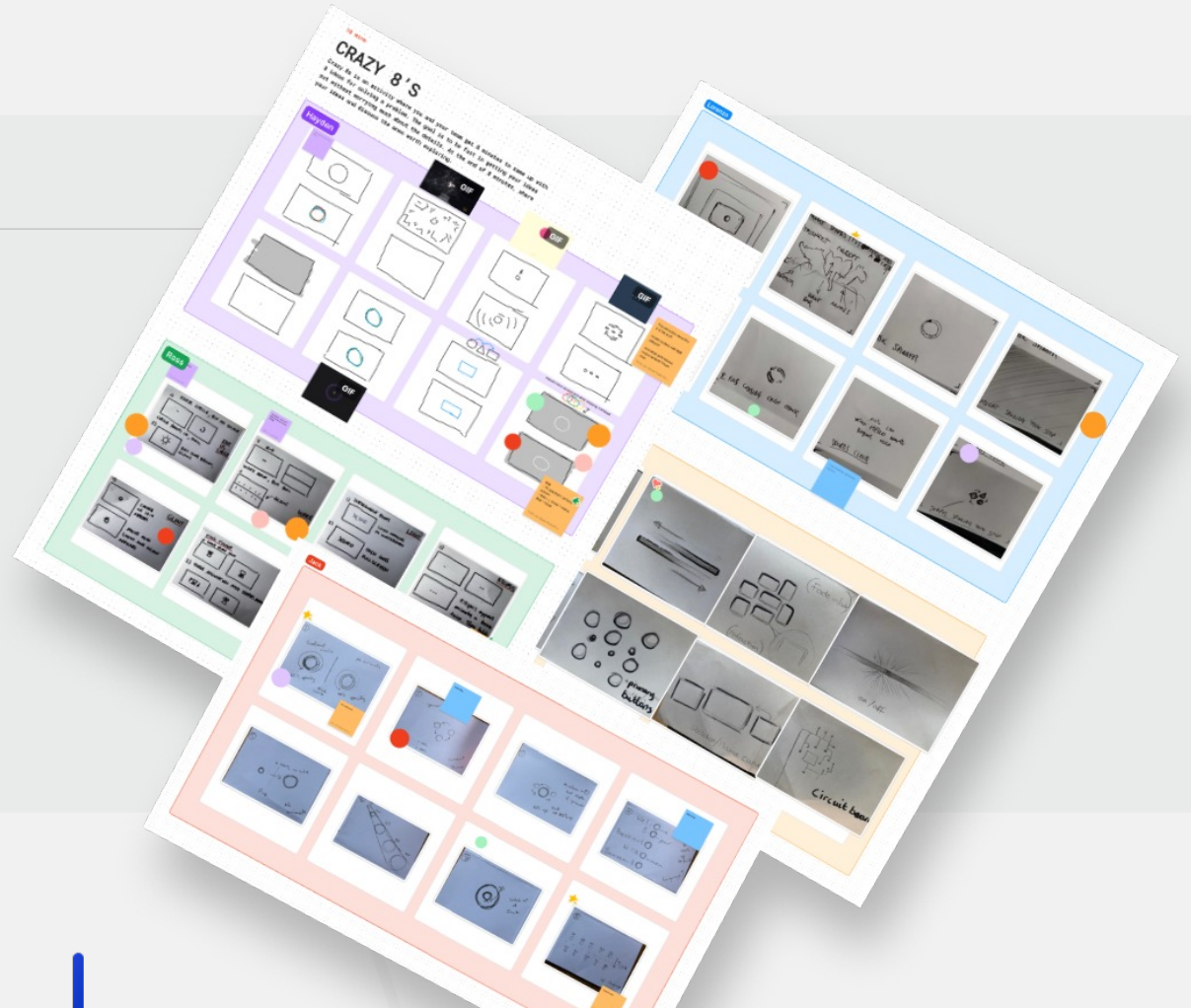
Crazy 8's

## Crazy 8's workshop

Ross, Lorenzo, Hayden, Jack, Amy.

1.5hr Sketching workshop.

[Link to the board](#)



# Concept Design

## Initial boot

### EntOS Circle/V2

Leaning on the existing Ent OS identity but using it in a way that is abstracted away enough that it is tied to it for those that know the brand but ultimately neutral enough to not be for end users.

### RGB

Using the 3 primary colours of the light spectrum (and coincidentally the 3 colours we use on the remote) to show how we form everything the user is about to see.

### Ellipsis

Typically, the Ellipsis is used to communicate 'more', it appears in our system in various places. This could be a playful and effective way to communicate the bootloader.

### Wake

Using the existing wake pattern but using neutral colours. On first boot the device doesn't use this animation already so we could introduce it here without it being a repetition, as long as it is fully replaced on subsequent boots.

### Edge

Playing on our Z-axis system transition, we could animate in giving the sense of immersion.

### Languages

The idea is that the bootloader embodies our truly global ethos with the use of text/ languages.

# Concept Design

Subsequent boot

xumo

 ROGERS

sky

# Next steps

## Timeline

1

Friday 14<sup>th</sup> June:  
**Design refinement**

2

Monday 17<sup>th</sup> June:  
**DBE Final Review**

3

Tuesday 18<sup>th</sup> June:  
**UI assets/specs**

4

Wednesday 19<sup>th</sup> June:  
**Dev handover**

Friday 28<sup>th</sup> June (*date TBC*): **Completion**



Thank you